

lab 1

SELECT

```
    relname AS object_name,  
    pg_size_pretty(pg_table_size(fed.oid)) AS data_size,  
    pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,  
    pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
```

FROM pg_class fed

JOIN pg_namespace n ON n.oid = fed.relnamespace

WHERE fed.relkind = 'r' AND n.nspname = 'public'

ORDER BY pg_total_relation_size(fed.oid) DESC;

The screenshot shows a PostgreSQL client interface with a query editor and a results pane. The query editor contains the following SQL code:

```
-- =====  
-- LAB 1: Inspecting Table Size  
-- Mục tiêu: Xác định bảng nào trong dvdrental "Index Heavy"  
-- =====  
  
SELECT  
    relname AS object_name,  
    pg_size_pretty(pg_table_size(fed.oid)) AS data_size,  
    pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,  
    pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size  
FROM pg_class fed  
JOIN pg_namespace n ON n.oid = fed.relnamespace  
WHERE fed.relkind = 'r' AND n.nspname = 'public'  
ORDER BY pg_total_relation_size(fed.oid) DESC;
```

The results pane displays the following table:

	object_name	data_size	index_size	total_size
1	rental	1624 kB	1120 kB	2744 kB
2	payment	1048 kB	920 kB	1968 kB
3	film	736 kB	200 kB	936 kB
4	film_actor	408 kB	216 kB	624 kB
5	inventory	408 kB	208 kB	616 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	city	64 kB	48 kB	112 kB
9	film_category	72 kB	40 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB

Total rows: 15 Query complete 00:00:00.118

lab 2

-- Bước 1: Kích thước index của bảng payment TRƯỚC khi tạo

SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_before;

-- Bước 2: Tạo index mới (multicolumn B-Tree trên 3 cột)

-- Nếu chạy lại nhiều lần, xóa trước để tránh lỗi trùng tên:

DROP INDEX IF EXISTS idx_payment_date_long;

CREATE INDEX idx_payment_date_long

ON payment (payment_date, amount, customer_id);

-- Bước 3: Kích thước SAU khi tạo

SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_after;

-- Bước 4: Đo trực tiếp index vừa tạo (chính xác, đơn giản nhất)

SELECT

pg_size_pretty(pg_relation_size('idx_payment_date_long')) AS index_size,

pg_relation_size('idx_payment_date_long') / 8192 AS pages_used;

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Dashboard x dvdrental/postgre... x dvdrental/postgre... x dvdrental/postgres@PostgreSQL 16* x

dvdrental/postgres@PostgreSQL 16

Query

Query History

Execute script FS

```
1 SELECT
2     relname AS object_name,
3     pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
4     pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
5     pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
6 FROM pg_class fed
7 JOIN pg_namespace n ON n.oid = fed.relnamespace
8 WHERE fed.relkind = 'r' AND n.nspname = 'public'
9 ORDER BY pg_total_relation_size(fed.oid) DESC;
10
11 -- Bước 1: Kích thước index của bảng payment TRƯỚC khi tạo
12 SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_before;
13
14
```

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1 of 1

size_before	text
1	920 kB

✓ Successfully run. Total query runtime: 74 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.074 LF Ln 14, Col 1

Object Explorer

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Dashboard X dvrental/postgres@PostgreSQL 16 X dvrental/postgres@PostgreSQL 16+ X

dvrental/postgres@PostgreSQL 16

Query Query History

```
1 SELECT
2     relname AS object_name,
3     pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
4     pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
5     pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
6 FROM pg_class fed
7 JOIN pg_namespace n ON n.oid = fed.relnamespace
8 WHERE fed.relkind = 'r' AND n.nspname = 'public'
9 ORDER BY pg_total_relation_size(fed.oid) DESC;
10
11 -- Bước 1: Kích thước index của bảng payment TRƯỚC khi tạo
12 SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_before;
13
14 -- Bước 2: Tạo index mới (multicolumn B-Tree trên 3 cột)
15 -- Nếu chạy lại nhiều lần, xóa trước để tránh lỗi trùng tên:
16 -- DROP INDEX IF EXISTS idx_payment_date_long;
17 CREATE INDEX idx_payment_date_long
18 ON payment (payment_date, amount, customer_id);
19
```

Execute script FS

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 60 msec.

Total rows: Query complete 00:00:00.060 LF Ln 18, Col 48

Object Explorer

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Dashboard X dvrental/postgres@PostgreSQL 16 X dvrental/postgres@PostgreSQL 16+ X

dvrental/postgres@PostgreSQL 16

Query Query History

```
5     pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
6 FROM pg_class fed
7 JOIN pg_namespace n ON n.oid = fed.relnamespace
8 WHERE fed.relkind = 'r' AND n.nspname = 'public'
9 ORDER BY pg_total_relation_size(fed.oid) DESC;
10
11 -- Bước 1: Kích thước index của bảng payment TRƯỚC khi tạo
12 SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_before;
13
14 -- Bước 2: Tạo index mới (multicolumn B-Tree trên 3 cột)
15 -- Nếu chạy lại nhiều lần, xóa trước để tránh lỗi trùng tên:
16 DROP INDEX IF EXISTS idx_payment_date_long;
17 CREATE INDEX idx_payment_date_long
18 ON payment (payment_date, amount, customer_id);
19
20 -- Bước 3: Kích thước SAU khi tạo
21 SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_after;
22
```

Execute script FS

Data Output Messages Notifications

	size_after text
1	1488 kB

Showing rows: 1 to 1 Page No: 1 of 1

Successfully run. Total query runtime: 63 msec. 1 rows affected. X

Total rows: 1 Query complete 00:00:00.063 LF Ln 22, Col 1

The screenshot shows a PostgreSQL IDE interface. The left sidebar displays a tree view of the database structure, including servers, databases, and schemas. The main window is divided into several panes: a Query pane with SQL code, a Query History pane, a Data Output pane, and a Messages/Notifications pane. The SQL code in the Query pane is as follows:

```

8 WHERE fed.relkind = 'r' AND n.nspname = 'public'
9 ORDER BY pg_total_relation_size(fed.oid) DESC;
10
11 -- Bước 1: Kích thước index của bảng payment TRƯỚC khi tạo
12 SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_before;
13
14 -- Bước 2: Tạo index mới (multicolumn B-Tree trên 3 cột)
15 -- Nếu chạy lại nhiều lần, xóa trước để tránh lỗi trùng tên:
16 DROP INDEX IF EXISTS idx_payment_date_long;
17 CREATE INDEX idx_payment_date_long
18 ON payment (payment_date, amount, customer_id);
19
20 -- Bước 3: Kích thước SAU khi tạo
21 SELECT pg_size_pretty(pg_indexes_size('payment')) AS size_after;
22 -- Bước 4: Đo trực tiếp index vừa tạo (chính xác, đơn giản nhất)
23 SELECT
24   pg_size_pretty(pg_relation_size('idx_payment_date_long')) AS index_size,
25   pg_relation_size('idx_payment_date_long') / 8192 AS pages_used;

```

The Data Output pane shows the results of the last query (line 25):

	index_size text	pages_used bigint
1	568 kB	71

The Messages pane at the bottom indicates: "Successfully run. Total query runtime: 49 msec. 1 rows affected."

lab 3

-- Tạo Partial Index (nếu chạy lại, xóa trước để tránh lỗi trùng tên)

```
DROP INDEX IF EXISTS idx_expensive_films;
```

```
CREATE INDEX idx_expensive_films
```

```
ON film (replacement_cost)
```

```
WHERE replacement_cost > 25.00;
```

-- Kiểm tra kích thước Partial Index vừa tạo

```
SELECT
```

```
    relname,
```

```
    pg_size_pretty(pg_relation_size(relid)) AS index_size
```

```
FROM pg_stat_user_indexes
```

```
WHERE relname = 'idx_expensive_films';
```

-- So sánh: tạo thử 1 Full Index (không WHERE) trên cùng cột để đối chiếu

-- Lưu ý: đặt tên khác để không trùng với idx_expensive_films

```
DROP INDEX IF EXISTS idx_full_replacement_cost;
```

```
CREATE INDEX idx_full_replacement_cost
```

```
ON film (replacement_cost);
```

```
SELECT
```

```
    relname,
```

```
    pg_size_pretty(pg_relation_size(relid)) AS index_size
```

```
FROM pg_stat_user_indexes
```

```
WHERE relname IN ('idx_expensive_films', 'idx_full_replacement_cost');
```

--Dọn dẹp sau khi so sánh xong (tùy chọn, nếu không cần giữ lại)

```
DROP INDEX idx_full_replacement_cost;
```

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dvdrental/postgres@PostgreSQL 16

Query Query History

37 WHERE relname = 'idx_expensive_films';
38
39 -- So sánh: tạo thử 1 Full Index (không WHERE) trên cùng cột để đối chiếu
40 -- Lưu ý: đặt tên khác để không trùng với idx_expensive_films
41 DROP INDEX IF EXISTS idx_full_replacement_cost;
42 CREATE INDEX idx_full_replacement_cost
43 ON film (replacement_cost);
44
45
46 SELECT
47 relname,
48 pg_size_pretty(pg_relation_size(relid)) AS index_size
49 FROM pg_stat_user_indexes
50 WHERE relname IN ('idx_expensive_films', 'idx_full_replacement_cost');
51 --Dọn dẹp sau khi so sánh xong (tùy chọn, nếu không cần giữ lại)
52 DROP INDEX idx_full_replacement_cost;
53
54

Scratch Pad x

Data Output Messages Notifications

DROP INDEX

Query returned successfully in 62 msec.

Total rows: Query complete 00:00:00.062

Ln 51, Col 20

Query returned successfully in 62 msec. X